



Solutions:

1. B. $\begin{pmatrix} -4 & -1 & -11 \\ 8 & -1 & 2 \end{pmatrix}$

$$\begin{pmatrix} 1 & 2 & -3 \\ 2 & 1 & 3 \end{pmatrix} - x = \begin{pmatrix} 5 & 1 & 8 \\ -6 & 0 & 5 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 & 2 & -3 \\ 2 & 1 & 3 \end{pmatrix} - \begin{pmatrix} 5 & 1 & 8 \\ -6 & 0 & 5 \end{pmatrix}$$

Calculating:

$$x = \begin{pmatrix} -4 & -1 & -11 \\ 8 & -1 & 2 \end{pmatrix}$$

2. C. $\begin{bmatrix} -3 & -5 \\ 4 & 1 \end{bmatrix}$

$$C_{11} = +(-3) = -3$$

$$C_{12} = -(5) = -5$$

$$C_{21} = -(-4) = 4$$

$$C_{22} = +(1) = 1$$

$$\begin{bmatrix} -3 & -5 \\ 4 & 1 \end{bmatrix} \rightarrow \text{Ans}$$

3. A. $\begin{bmatrix} 2 \\ -1 \end{bmatrix}^*$

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = 0$$

$$\begin{vmatrix} 2-\lambda & 2 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(3-\lambda) - 2 = 0$$

$$\lambda^2 - 5\lambda + 6 - 2 = 0$$

$$\lambda^2 - 5\lambda + 4 = 0$$

$$\lambda_1 = 1$$

$$\lambda_2 = 4$$

Then at $\lambda = 1$

$$\begin{bmatrix} 2-\lambda & 2 \\ 1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x+2y \\ x+2y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x + 2y = 0$$

All (x, y) that can satisfy the equation $x + 2y = 0$ is/are the correct answer.

4. D. $x^3 + x = 0$

roots = -i, i and 0

$$x(x+i)(x-i) = 0$$

$$x(x^2 - ix + ix + 1) = 0$$

$$x(x^2 + 1) = 0$$

$$x^3 + x = 0$$

5. C. 2

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

$$R2 - R1 \times 1 \text{ and } R3 - R1 \times 2:$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix}$$

$$R3 - R2 \times 1:$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of matrix is number of independent rows or columns.
 Since, one row has elements all zero, only 2 rows are independent.
 Therefore, rank of matrix is 2.

6. C. $\frac{\omega}{s^2 + \omega^2}$

$$L\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}$$

Reverse Engineering:

$$L^{-1}\left\{\frac{1}{s^2 + s}\right\} = L^{-1}\left\{\frac{1}{s} - \frac{1}{s+1}\right\} = 1 - e^{-t}$$

Reverse Engineering:

Given: Let $s = 5$ and $\omega = 1$

$$\int_0^\infty f(x) e^{-st} dt = \int_0^8 (\sin t) e^{-5t} dt = \frac{1}{26}$$

From choices:

$$\frac{\omega}{s^2 + \omega^2} = \frac{1}{5^2 + 1} = \frac{1}{26}$$

7. C. $-\frac{2}{5} + \frac{21}{5}i$

Convert to polar form:
 $8 + 5i$:

$$r = \sqrt{8^2 + 5^2} = \sqrt{89}$$

$$\theta = \tan^{-1} \frac{5}{8} = 32^\circ$$

$1 - 2i$:

$$r = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$\theta = \tan^{-1} \frac{-2}{1} = -63.44^\circ$$

So,

$$\frac{\sqrt{89} \angle 32^\circ}{\sqrt{5} \angle -63.435^\circ} = \frac{\sqrt{89}}{\sqrt{5}} \angle (32^\circ + 63.435^\circ) = \frac{\sqrt{445}}{5} \angle 95.44^\circ$$

Convert back to rectangular form:

$$x = r \cos \theta$$

$$x = \frac{\sqrt{445}}{5} \cos 95.44^\circ = -\frac{2}{5}$$

$$y = r \sin \theta$$

$$y = \frac{\sqrt{445}}{5} \sin 95.44^\circ = \frac{21}{5}$$

$$x + yi = -\frac{2}{5} + \frac{21}{5}i$$



Calculator Shortcut: (Using Casio FX570ES+)

Use calculator in COMPLEX MODE

$$\frac{8+5i}{1-2i} = -\frac{2}{5} + \frac{21}{5}i$$

8. A. 0

$$\binom{n}{0} = 1 \text{ for any value of } n$$

So, the expression will only be equal if 'x' is zero.

$$\binom{6}{x} = \binom{4}{x} \rightarrow 1 = 1$$

9. D. z^{-5}

$$\delta[n-5] \rightarrow \sum_{n=-\infty}^{\infty} \delta[n-5]z^{-n} = \delta[5-5]z^{-5} = z^{-5}$$

Region of convergence: all z, except z = 0.

10. A. $\begin{pmatrix} -12 & 30 \\ -11 & 0 \end{pmatrix}$

$$BA = \begin{pmatrix} 6 & 0 \\ 3 & -5 \end{pmatrix} \begin{pmatrix} -2 & 5 \\ 1 & 3 \end{pmatrix}$$

$$BA = \begin{pmatrix} 6(-2) + 0(1) & 6(5) + 0(3) \\ 3(-2) + (-5)(1) & 3(5) + (-5)(3) \end{pmatrix}$$

$$BA = \begin{pmatrix} -12 & 30 \\ -11 & 0 \end{pmatrix}$$

Calculator Shortcut: Using Casio FX570ES+, Input the given matrices in your calculator as Matrix A and B respectively.

Mode $\rightarrow$ 6: MATRIX $\rightarrow$ 1: MatA

$$\rightarrow 5: 2 \times 2 \rightarrow \begin{bmatrix} -2 & 5 \\ 1 & 3 \end{bmatrix} \rightarrow \boxed{\text{AC}}$$

Shift $\rightarrow$ [4] Matrix $\rightarrow$ 2: Data $\rightarrow$ 2: MatB

$$\rightarrow 5: 2 \times 2 \rightarrow \begin{bmatrix} 6 & 0 \\ 3 & -5 \end{bmatrix} \rightarrow \boxed{\text{AC}}$$

$$\text{MatB} \times \text{MatA} = \begin{bmatrix} -12 & 30 \\ -11 & 0 \end{bmatrix}$$

11. D. $2.88 - j0.492^*$

$$2 \frac{\ln(1+j\sqrt{3})}{\ln(1+j)} = 2 \frac{\ln 2 + j\frac{1}{3}\pi}{\ln \sqrt{2} + j\frac{1}{4}\pi} = 2.88 - j0.492$$

12. A. -i

$$\sum_{j=0}^{10} (-i)^j = 1 - i - 1 + i + 1 - i - 1 + i + 1 - i - 1 = -i$$

13. C. 1

For a projectile that is a full-parabola:

$$p(x) = \sum_{k=0}^n \frac{(x-x_0)^k}{k!} f^{(k)}(x_0)$$

From x, $x_0 = 0$, $k=1$ and $f(x) = e^{\sin x}$

Get the first derivative of $f(x)$.

$$f(x) = e^{\sin x} \Rightarrow f'(x) = (\cos x)e^{\sin x}$$

$$\text{at } x=0 \Rightarrow f'(0) = (\cos 0)e^{\sin 0} = 1$$

$$\text{So, at } k=1 = \frac{(x-0)^1}{1!} (1) = x$$

Coefficient is 1

14. A. $1 - \frac{1}{2}\left(x - \frac{\pi}{2}\right)^2 + \frac{1}{24}\left(x - \frac{\pi}{2}\right)^4$

$$p(x) = \sum_{k=0}^n \frac{(x-x_0)^k}{k!} f^{(k)}(x_0)$$

From, $x_0 = \frac{\pi}{2}$, $k=2$ and $f(x) = \sin x$

$$f(x) = \sin x, \quad f\left(\frac{\pi}{2}\right) = 1$$

$$f^1(x) = \cos x, \quad f^1\left(\frac{\pi}{2}\right) = 0$$

$$f^2(x) = -\sin x, \quad f^2\left(\frac{\pi}{2}\right) = -1$$

$$f^3(x) = -\cos x, \quad f^3\left(\frac{\pi}{2}\right) = 0$$

$$f^4(x) = \sin x, \quad f^4\left(\frac{\pi}{2}\right) = 1$$

$$p(x) = \frac{\left(x - \frac{\pi}{2}\right)^0}{0!} (1) + \frac{\left(x - \frac{\pi}{2}\right)^2}{2!} (-1) + \frac{\left(x - \frac{\pi}{2}\right)^4}{4!} (1)$$

$$p(x) = 1 - \frac{1}{2}\left(x - \frac{\pi}{2}\right)^2 + \frac{1}{24}\left(x - \frac{\pi}{2}\right)^4$$

15. A. $\begin{pmatrix} -9 & 5 \\ 2 & -1 \end{pmatrix}$

Solving the determinant of the given matrix:

$$D = \begin{vmatrix} 1 & 5 \\ 2 & 9 \end{vmatrix}$$

$$D = 9 - 10$$

$$D = -1$$

Note: For a 2 x 2 Matrix, say, matrix

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix}, \text{ its inverse is given by:}$$

$$A^{-1} = \frac{1}{D} \begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$$

Thus,

$$A^{-1} = \frac{1}{(-1)} \begin{vmatrix} 9 & -5 \\ -2 & 1 \end{vmatrix}$$

$$A^{-1} = \begin{vmatrix} -9 & 5 \\ 2 & -1 \end{vmatrix}$$

16. A. 3i

Since 20 is exactly divisible by 4,
 $i^{20} = 1$,

$$i^{21} = i$$

Since 28 is exactly divisible by 4,
 $i^{28} = 1$,

$$i^{29} = i$$

Substituting:

$$i^{29} + i^{21} + i = i + i + i = 3i$$

17. C. $2i - 1$

Since 3216 is exactly divisible by 4,
 $i^{3216} = 1$,

$$i^{3217} = i$$

Since 424 is exactly divisible by 4

$$i^{424} = 1,$$

$$i^{425} = i$$



$$i^{426} = -1$$

$$i^{427} = -i$$

Since 16 is exactly divisible by 4

$$i^{16} = 1$$

$$i^{17} = i$$

$$i^{18} = -1$$

Substituting:

$$i^{3217} - i^{427} + i^{18} = i - (-i) + (-1) = 2i - 1$$

18. A. $2 + 3x - x^2 - \frac{1}{2}x^3 + \dots$

The Taylor series of $\cos x$ is

$$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

The Taylor series of $\sin x$ is

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

So,

$$\begin{aligned} 3\sin x + 2\cos x &= 3\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right) \\ &\quad + 2\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) \\ &= 2 + 3x - x^2 - \frac{x^3}{2} + \dots \rightarrow \text{Ans} \end{aligned}$$

19. A. $\sec^2 x$

$$A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

$$\begin{aligned} |A| &= 1 + \tan^2 x \\ &= \sec^2 x \end{aligned}$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

$$\begin{aligned} A^T A^{-1} &= \frac{1}{|A|} \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} \\ &= \frac{1}{\sec^2 x} \begin{bmatrix} 1 - \tan^2 x & -2\tan x \\ 2\tan x & 1 - \tan^2 x \end{bmatrix} \end{aligned}$$

$$\begin{aligned} |A^T A^{-1}| &= \frac{1}{\sec^2 x} \left[(1 - \tan^2 x)^2 + 4\tan^2 x \right] \\ &= \frac{1}{\sec^2 x} [1 + \tan^4 x - 2\tan^2 x + 4\tan^2 x] \end{aligned}$$

$$= \frac{1}{\sec^2 x} [1 - \tan^4 x + 2\tan^2 x]$$

$$= \frac{1}{\sec^2 x} (1 - \tan^2 x)^2$$

$$= \frac{(\sec^2 x)^2}{\sec^2 x}$$

$$= \sec^2 x \rightarrow \text{Ans}$$

20. A. 1

One of the eigenvalues is zero. That implies that the determinant of the matrix is also zero. From the matrix A, we can see that for determinant to be zero, Row 1 and Row 3 can be made same.

$$\frac{-9+x}{2} = -4$$

$$x = 1 \rightarrow \text{Ans}$$

21. B. second

$$\begin{aligned} \frac{1+2i}{1-i} &= \frac{1+2i}{1-i} \times \frac{1+i}{1+i} \\ &= \frac{-1+3i}{2} \\ &= -\frac{1}{2} + \frac{3}{2}i \end{aligned}$$

This is represented by the point

$$\left(-\frac{1}{2}, \frac{3}{2}\right) \text{ in the second quadrant.}$$

22. D. $\frac{3}{16} - \frac{\sqrt{7}i}{16}$

$$\text{Reciprocal of } z = \frac{\bar{z}}{|z|^2}$$

Therefore the reciprocal of $3 + \sqrt{7}i$

$$= \frac{3 - \sqrt{7}i}{16} = \frac{3}{16} - \frac{\sqrt{7}i}{16}$$

23. D. $0 + \frac{3}{2}i$

$$z = \frac{\sqrt{5+12i} + \sqrt{5-12i}}{\sqrt{5+12i} - \sqrt{5-12i}} \times \frac{\sqrt{5+12i} + \sqrt{5-12i}}{\sqrt{5+12i} + \sqrt{5-12i}}$$

$$z = \frac{5+12i+5-12i+2\sqrt{25+144}}{5+12i-5+12i}$$

$$z = \frac{3}{2i} = \frac{3i}{-2} = 0 - \frac{3}{2}i$$

$$\text{Conjugate of } z = 0 + \frac{3}{2}i$$

24. B. $\cos \frac{\pi}{2} - i \sin \frac{\pi}{2}$

$$z = (i^{25})^3$$

$$= i^{75}$$

$$= i^{4 \times 18 + 3}$$

$$= (i^4)^{18} (i)^3$$

$$= i^3$$

$$= -i$$

$$z = 0 - i$$

Polar form:

$$z = r(\cos \theta + i \sin \theta)$$

$$z = 1 \left[\cos \left(-\frac{\pi}{2} \right) + i \sin \left(-\frac{\pi}{2} \right) \right]$$

$$z = \cos \frac{\pi}{2} - i \sin \frac{\pi}{2}$$

25. B. $1 - e^{-t}$

$$f(s) = L^{-1} \left[\frac{1}{(s^2 + s)} \right]$$

$$\frac{1}{(s^2 + s)} = \frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$$

$$L^{-1} \left[\frac{1}{(s^2 + s)} \right] = L^{-1} \left[\frac{1}{s} - \frac{1}{s+1} \right]$$

$$= L^{-1} \left[\frac{1}{s} \right] - L^{-1} \left[\frac{1}{s+1} \right]$$

$$= 1 - e^{-t}$$